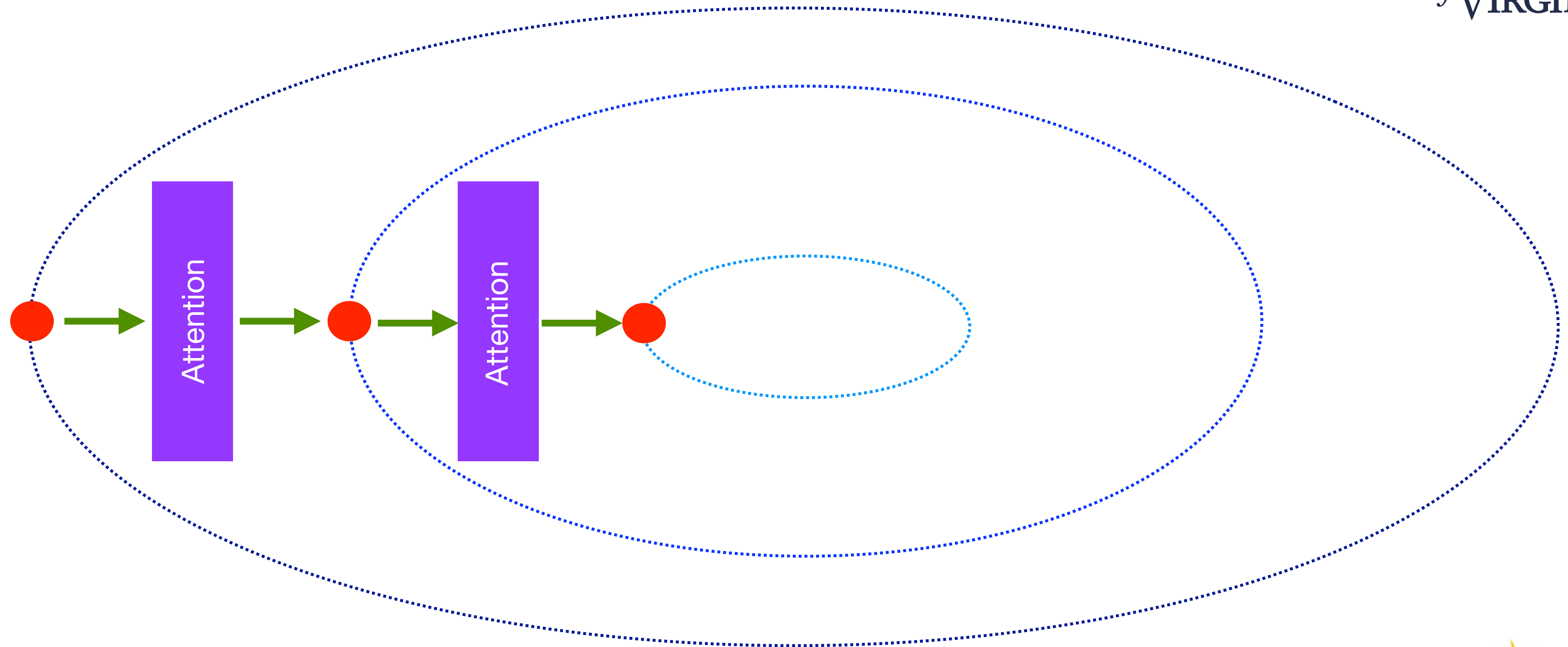


In-Context Learning for Discrete Optimal Transport: Can Transformers Sort?

Hadi Daneshmand



How do transformers translate?

OT Problem

- ▶ Optimal transport:

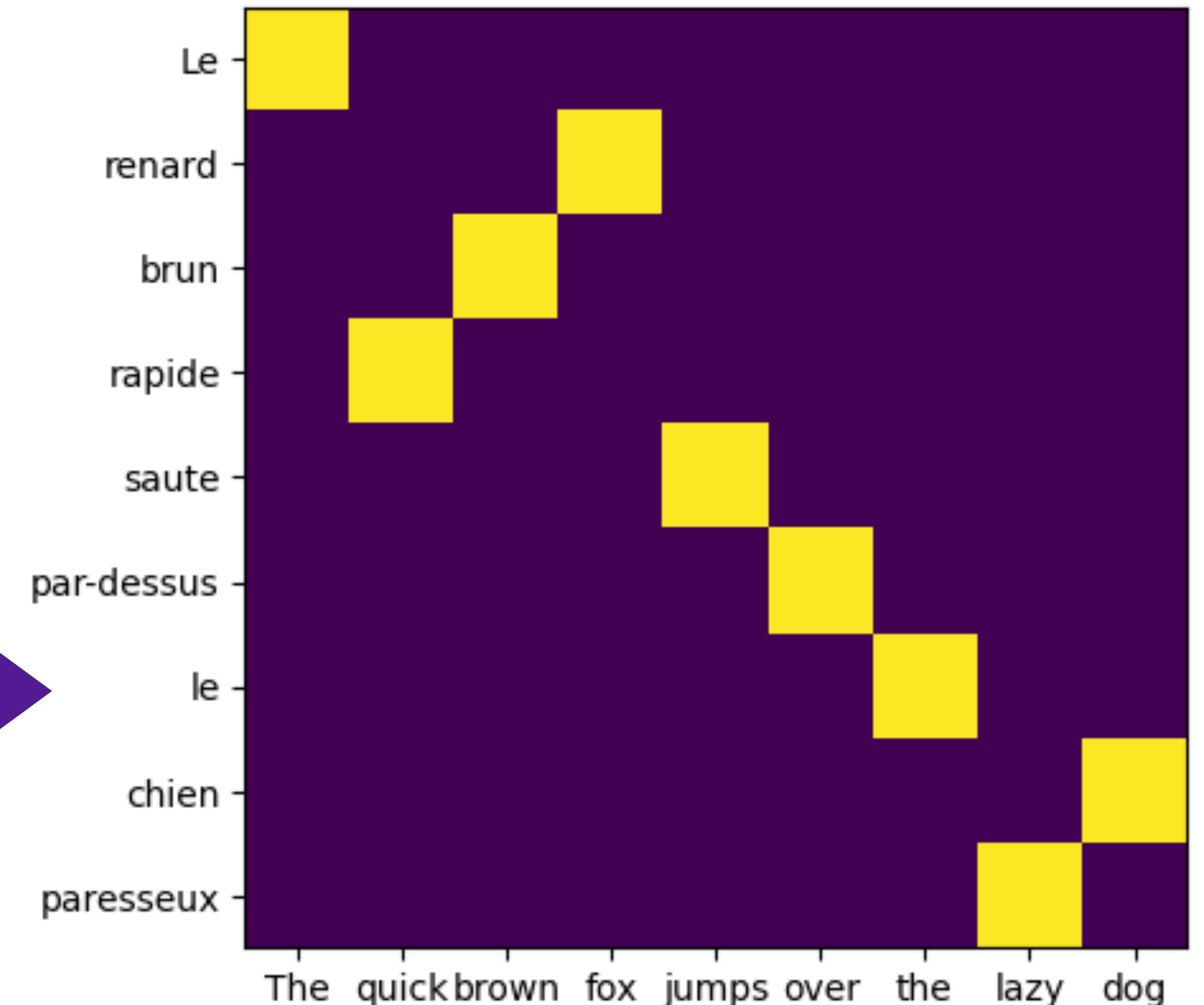
$$P^* := \arg \min_P \sum_{ij} \|x_i - y_j\|^2 P_{ij} \text{ subject to } P \text{ is a permutation matrix}$$

- ▶ Permutation matrix: $P \in \mathbb{R}^{n \times n}$, $P_{ij} \in \{0,1\}$, $\sum_{j=1}^n P_{ij} = 1$

- ▶ Applications: sorting, cell perturbation analysis, reviewer/paper matching, ...

4

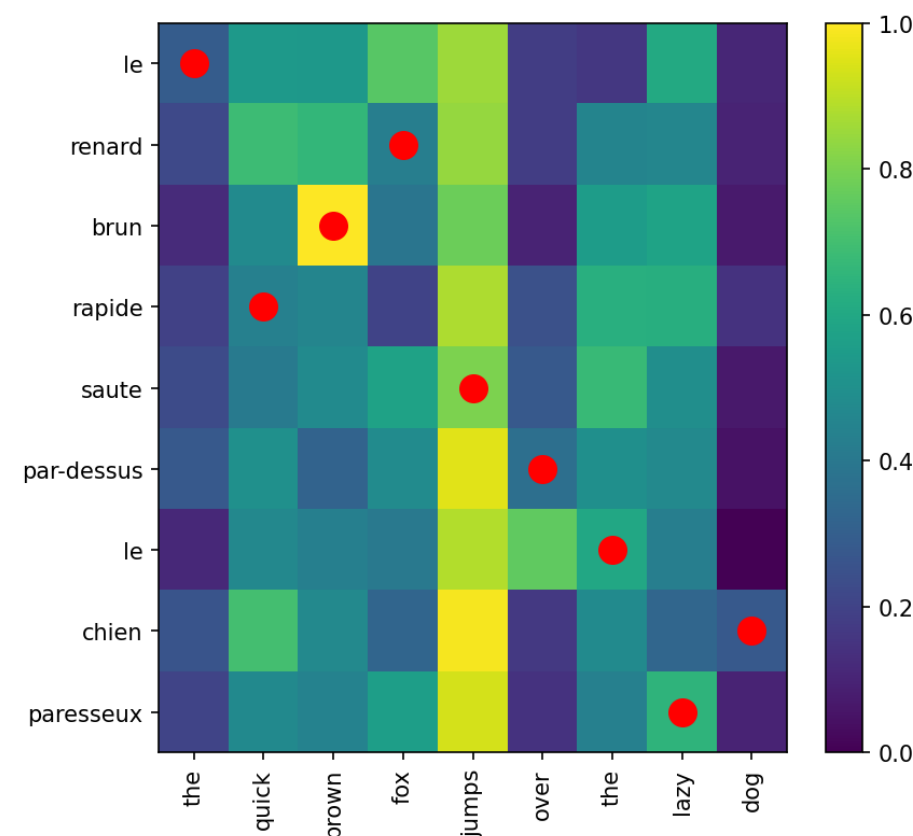
- $$P^* := \arg \min_P \sum_{ij} \|x_i - y_j\|^2 P_{ij} \text{ subject to } \dots \rightarrow$$



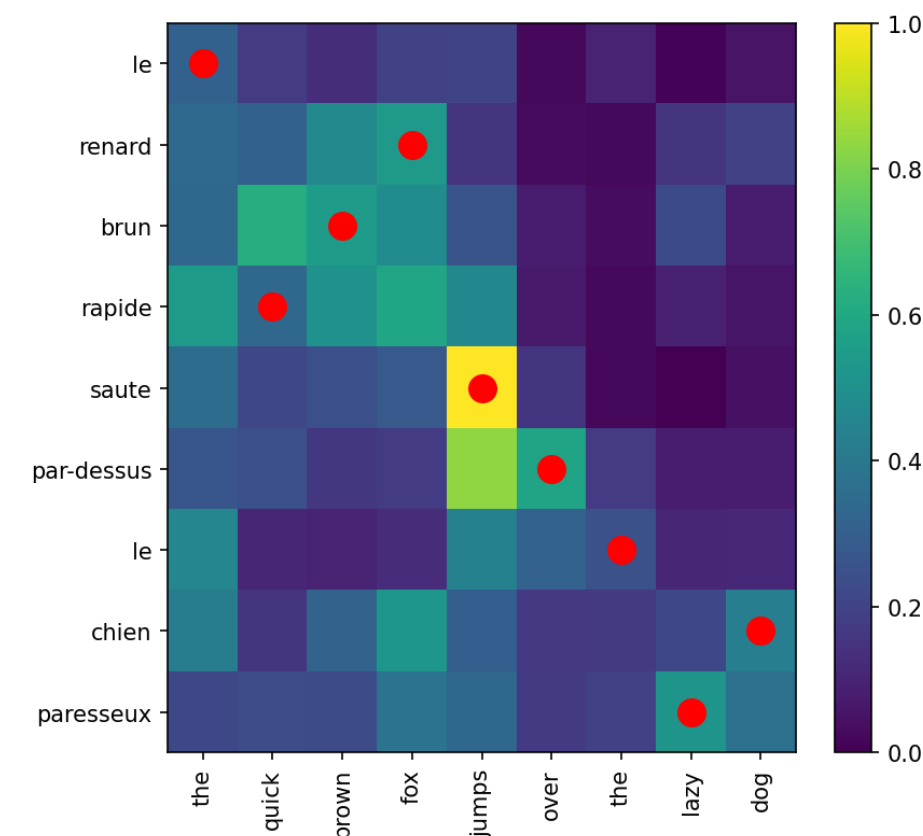
Attention, OT, translation

5

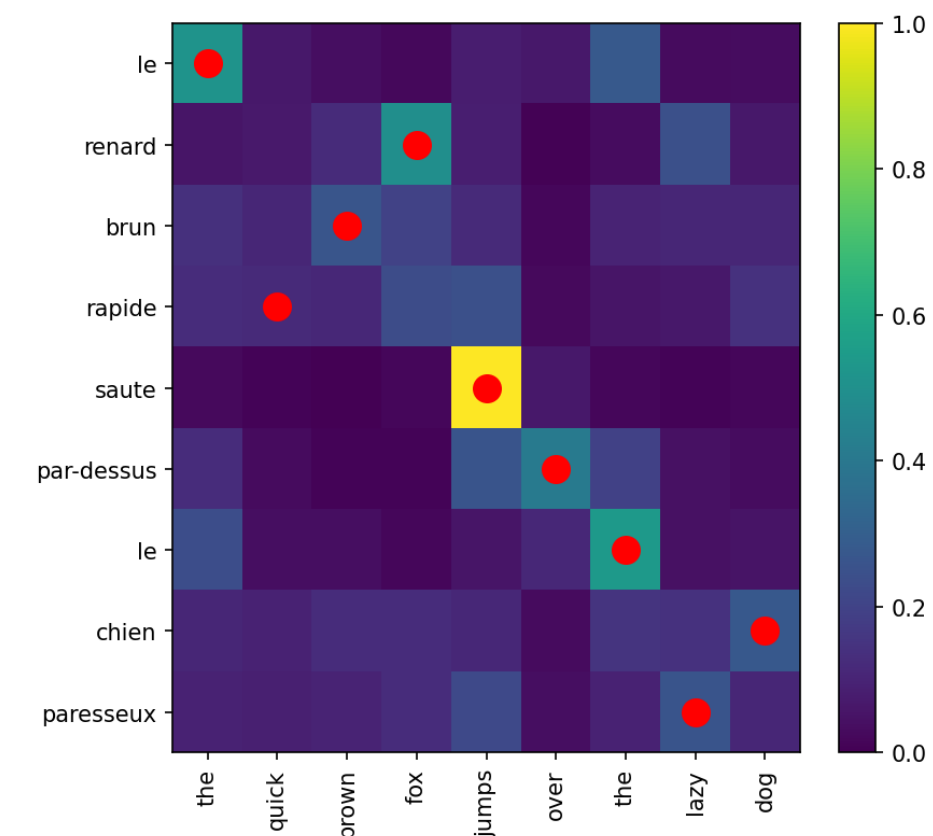
- ▶ Attention weights converge toward the OT solution across layers



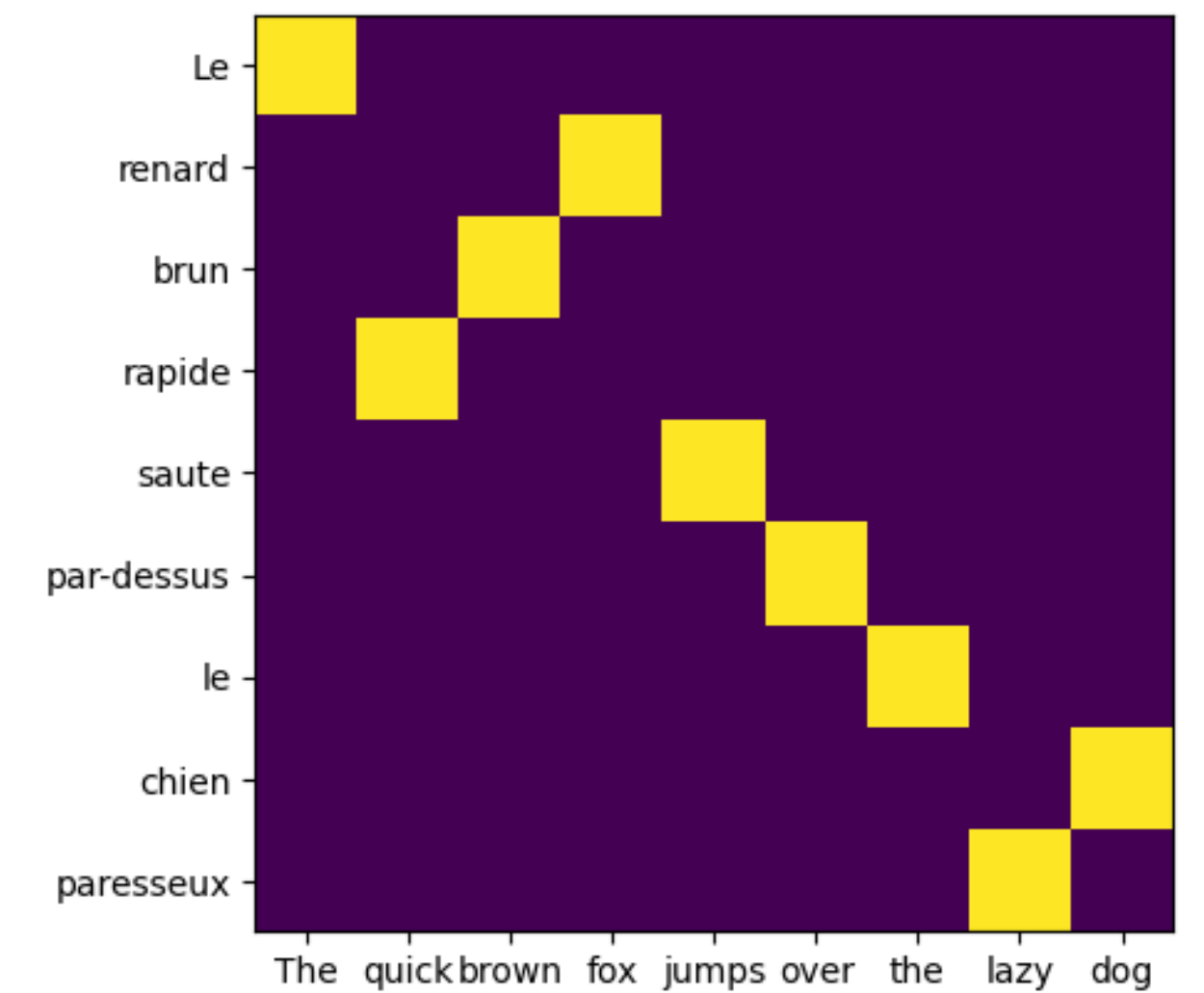
▶ Layer 1



▶ Layer 6



▶ Layer 12



▶ OT of embeddings

Main Result

Theorem (informal statement)

A transformer with ℓ softmax attention layers can approximate OT up to error

$$\frac{\sqrt{n}}{\sqrt{\ell}} \times \text{data dependent and regularization constants}$$

Dual objective of OT

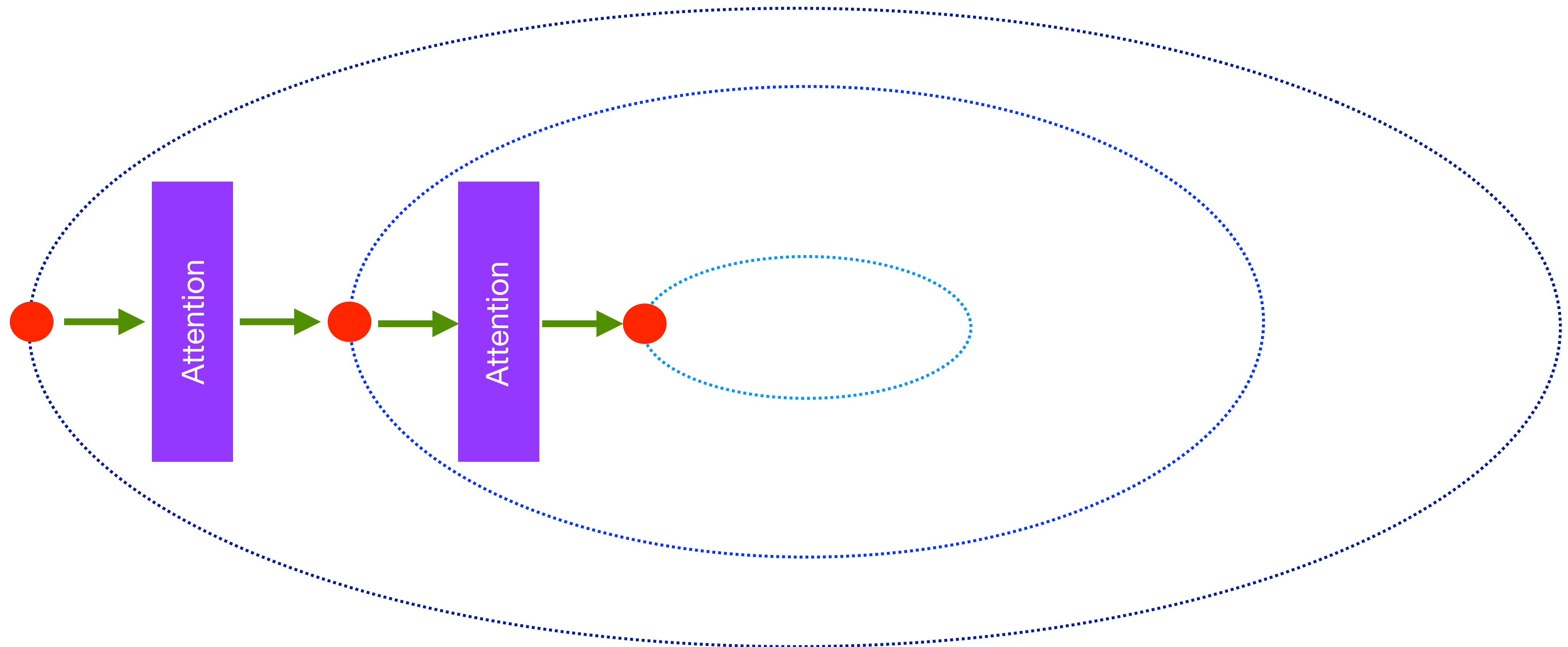
7

Entropy Regularized OT: $P_\lambda = \arg \min_P \sum_{ij} \|x_i - y_j\|^2 P_{ij} + \lambda \underbrace{\sum_i \sum_j P_{ij} \log(P_{ij})}_{-entropy}$

Dual: $\min_{u_i, v_j} L(u, v) = \lambda \sum_{ij} e^{\frac{-\|x_i - y_j\|^2 + u_i + v_j}{\lambda}} - 1 - \sum_i v_i - \sum_j u_j$

Mechanism of Attention for Translation

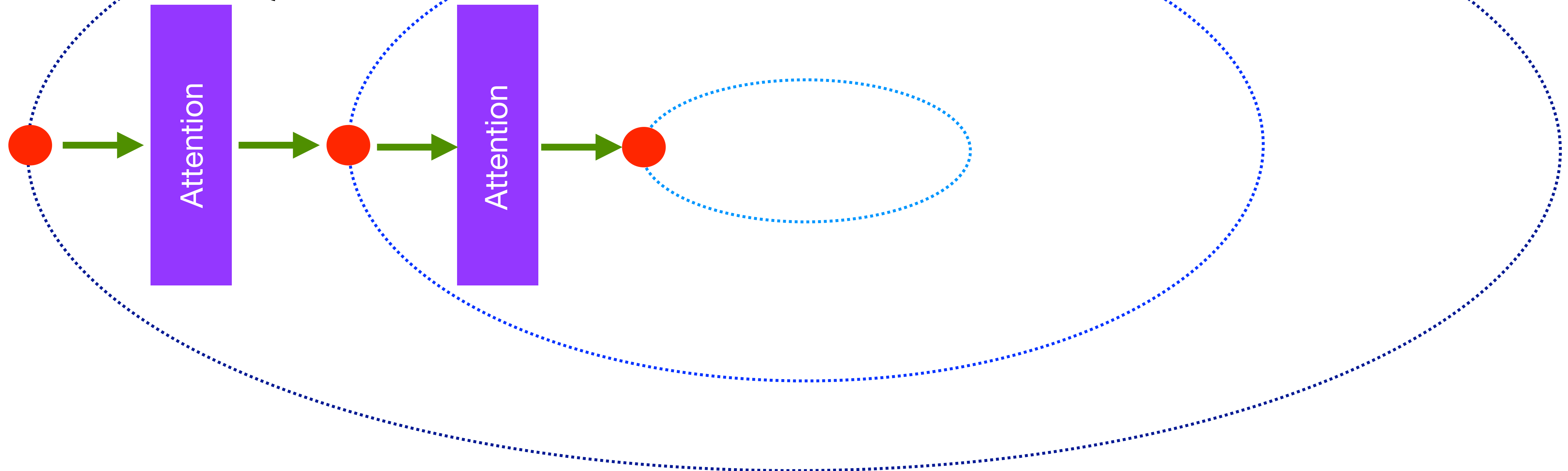
- ▶ Attention layers can simulate GD with adaptive step sizes on the dual function L



Mechanism of Attention for Translation

► Dual: $\min_{u_i, v_j} L(u, v) = \lambda \sum_{ij} e^{\frac{-\|x_i - y_j\|^2 + u_i + v_j}{\lambda}} - 1 - \sum_i v_i - \sum_j u_j$

$$\begin{cases} u_0 = 0 \\ v_0 = 0 \end{cases} \quad \begin{cases} u_1 = u_0 - \text{diag}(d_0) \nabla_u L(u_0, v_0) \\ v_1 = v_0 - \text{diag}(d'_0) \nabla_v L(u_0, v_0) \end{cases}$$



Related works

- ▶ Results on linear attention (not softmax):
 - Linear regression: [Ahn et al., 2023, Von Oswald et al., 2023]
 - Policy evaluation for RL [Wang et al., 2025]
 - Low-rank matrix completion [Lutz et al., 2025]
- ▶ Mechanistic analysis of In-context learning [Garg et al., 2022, Akyürek et al., 2023]

Thank you very much for your attention



- ▶ Website: <https://hadidaneshmand.github.io/papers/aistats26.html>
- ▶ Github: <https://github.com/hadidaneshmand/aistats26-incontextOT>
- ▶ Contact: dhadi at virginia dot edu